

Renal Reference — Acid-Base (Sketchy pHun Park)



1. Water pHun Park

WHAT YOU SEE

Sign: 'Water pHun Park'

WHAT IT ENCODES

The whole scene encodes systematic ABG interpretation. Work it in a fixed order every time: (1) pH — acidemic (<7.35) or alkalemic (>7.45)? (2) Is the primary process respiratory (PCO₂, normal 35-45) or metabolic (HCO₃, normal 22-28)? (3) Calculate the anion gap = Na – (Cl + HCO₃); normal ~8-12. (4) Check compensation. (5) If high gap, run delta-delta to unmask a second metabolic disorder. Mnemonic 'ROME': Respiratory Opposite (pH and PCO₂ move opposite directions), Metabolic Equal (pH and HCO₃ move same direction).

BOARD TRAP

WRONG answer is to read the pH alone and stop. A 'normal' pH of 7.40 can hide a mixed disorder (e.g., metabolic acidosis + respiratory alkalosis in salicylate toxicity). Always compute the anion gap and check compensation even when pH looks normal — the discriminator is that PCO₂ and HCO₃ are both abnormal in opposite directions.

2. Lemonade stand (acid)

WHAT YOU SEE

A bright 'Lemonade' stand with sour lemons on its sign — sour lemonade = the ACID side of the balance, i.e. acidemia (pH <7.35).

WHAT IT ENCODES

Lemonade = sour/acidic = the ACID side of the balance, i.e. acidemia (arterial pH <7.35). Acid is added either by retaining CO₂ (respiratory) or by gaining fixed acid / losing HCO₃ (metabolic). High-anion-gap metabolic acidosis causes are captured by MUDPILES (Methanol, Uremia, DKA, Propylene glycol, Iron/INH, Lactate, Ethylene glycol, Salicylates); normal-gap (hyperchloremic) causes by USED CARP (Ureterosigmoidostomy, Saline, Endocrine/Addison, Diarrhea, Carbonic anhydrase inhibitors, Ammonium chloride, RTA, Pancreatic fistula).

BOARD TRAP

WRONG answer is calling every metabolic acidosis 'high gap.' A normal-gap (hyperchloremic) acidosis from diarrhea or type 1/2 RTA has a NORMAL anion gap because Cl rises to replace lost HCO₃. Discriminator: compute the gap — diarrhea/RTA give normal gap; lactate/ketones/toxic alcohols give a high gap.

3. Tickets cone / pH meter

WHAT YOU SEE

Central rainbow 'TICKETS' cone over a tank reading 22-28

WHAT IT ENCODES

The central pH/HCO₃ reference gauge — color scale = the pH spectrum from acid (low) to base (high). Conceptually this is the Henderson-Hasselbalch relationship: $\text{pH} \propto \text{HCO}_3 / \text{PCO}_2$. Raising HCO₃ or lowering PCO₂ raises pH; the kidney controls the numerator (HCO₃, slow, days) and the lungs control the denominator (PCO₂, fast, minutes).

BOARD TRAP

WRONG answer is assuming the same absolute change in HCO₃ and PCO₂ affects pH equally. Because pH tracks the RATIO, what matters is proportional change — a small PCO₂ swing from rapid breathing can shift pH faster than a slow renal HCO₃ adjustment. Discriminator: respiratory changes are immediate; renal compensation lags 3-5 days.

4. Tank reading 22-28

WHAT YOU SEE

Numbers '22-28' inside the tank

WHAT IT ENCODES

Normal serum bicarbonate (HCO₃) range is 22-28 mEq/L — the metabolic arm of acid-base. HCO₃ <22 = metabolic acidosis; HCO₃ >28 = metabolic alkalosis. In a primary metabolic acidosis, respiratory compensation is predicted by Winter's formula: expected PCO₂ = 1.5 × HCO₃ + 8 (±2). If measured PCO₂ is higher than predicted there is a concurrent respiratory acidosis; if lower, a concurrent respiratory alkalosis.

BOARD TRAP

WRONG answer is treating a low HCO₃ as proof of metabolic acidosis without checking PCO₂. A low HCO₃ is ALSO the renal compensation for a chronic respiratory alkalosis (e.g., high-altitude, pregnancy). Discriminator: in primary metabolic acidosis pH is low; in compensated respiratory alkalosis pH is high or near-normal.

5. Normal pH hours 7.35-7.45

WHAT YOU SEE

'Normal pH hours 7:35 am to 7:45 am'

WHAT IT ENCODES

Normal arterial pH is 7.35-7.45 (clock = the pH 'opening hours'). pH <7.35 = acidemia; pH >7.45 = alkalemia. The terms acid-OSIS / alkal-OSIS describe the underlying process (which can coexist), while acid-EMIA / alkal-EMIA describe the net blood pH. Life-threatening thresholds: pH <7.1 risks arrhythmia and depressed contractility; pH >7.55 risks seizures, tetany, and arrhythmia.

BOARD TRAP

WRONG answer is to label a pH of exactly 7.40 as 'no acid-base disorder.' Mixed disorders (e.g., concurrent metabolic acidosis and metabolic alkalosis in a vomiting DKA patient) can produce a perfectly normal pH. Discriminator: an abnormal anion gap or a delta-delta ratio outside 1-2 reveals the hidden disorder despite a normal pH.

6. Smoking BBQ / grills

WHAT YOU SEE

Smoking grills / coils emitting smoke

WHAT IT ENCODES

Retained CO₂ (smoke) = respiratory acidosis from hypoventilation. Causes: COPD/asthma, opioid or benzodiazepine overdose (CNS depression), neuromuscular disease (Guillain-Barré, myasthenia), obesity hypoventilation. Acute: every 10 mmHg rise in PCO₂ raises HCO₃ by ~1 (and drops pH by ~0.08). Chronic: kidneys retain HCO₃ ~3.5 per 10 mmHg PCO₂ (pH drops only ~0.03). Treat the cause; ventilatory support (BiPAP) if needed.

BOARD TRAP

WRONG answer is giving supplemental O₂ freely to a hypercapnic COPD patient and ignoring rising CO₂. High-flow O₂ can blunt hypoxic drive and worsen CO₂ retention/narcosis. Discriminator: in chronic respiratory acidosis HCO₃ is markedly elevated (renal compensation) with near-normal pH — target controlled O₂ (SpO₂ 88-92%) and consider BiPAP.

7. Hyperventilating swimmer (resp alkalosis)

WHAT YOU SEE

Figure breathing hard / blowing on the left

WHAT IT ENCODES

Hyperventilation blows off CO₂ = respiratory alkalosis (high pH, low PCO₂). Causes: anxiety/pain, hypoxia (PE, pneumonia, high altitude), sepsis (early), pregnancy (progesterone), salicylate toxicity (direct respiratory center stimulation), liver failure. Compensation: acute HCO₃ falls ~2 per 10 mmHg drop in PCO₂; chronic ~4-5 per 10 mmHg. Symptoms: perioral/digital paresthesias, lightheadedness, carpopedal spasm (from reduced ionized calcium).

BOARD TRAP

WRONG answer is attributing tachypnea with low PCO₂ in a sick patient purely to 'anxiety.' Early sepsis, PE, and salicylate toxicity all present with primary respiratory alkalosis. Discriminator: salicylate poisoning classically gives a MIXED respiratory alkalosis + high-anion-gap metabolic acidosis — check the gap and a salicylate level before calling it panic.

8. Lifeguard on tower (compensation)

WHAT YOU SEE

Lifeguard seated on the tall chair watching over

WHAT IT ENCODES

Compensation = the secondary system 'watching' and adjusting to pull pH back toward normal (lungs respond to metabolic problems; kidneys respond to respiratory problems). Use formulas to test adequacy: Winter's for metabolic acidosis (expected PCO₂ = 1.5×HCO₃ + 8 ±2); for metabolic alkalosis expected PCO₂ = 0.7×HCO₃ + 21; respiratory rules above. A measured value off the predicted = a second (mixed) disorder.

BOARD TRAP

WRONG answer is assuming compensation can fully normalize the pH. Compensation NEVER fully corrects pH back to 7.40 and never overshoots to the opposite side. Discriminator: if the pH is normal or has crossed to the other side of 7.40, there is a second primary disorder, not 'complete compensation.'

9. Pool (the body fluid / pH bath)

WHAT YOU SEE

The large blue swimming pool on the right — the body's extracellular fluid 'bath' whose pH is defended by the bicarbonate buffer system.

WHAT IT ENCODES

The extracellular fluid compartment where pH is defended. The bicarbonate-carbonic acid system is the dominant ECF buffer; intracellular phosphate, proteins (hemoglobin), and bone provide additional buffering. In chronic metabolic acidosis (e.g., CKD, RTA) bone carbonate is mobilized as buffer, contributing to renal osteodystrophy and growth failure in children.

BOARD TRAP

WRONG answer is thinking the bicarbonate buffer can neutralize acid indefinitely without renal/respiratory help. Buffers only blunt the initial swing; they are consumed and must be regenerated — the kidney regenerates HCO₃ by excreting NH₄⁺ and titratable acid. Discriminator: failure to make new bicarbonate (e.g., distal/type 1 RTA, CKD) produces a persistent metabolic acidosis despite buffering.

10. Balloons / party (buffers)

WHAT YOU SEE

A cluster of party balloons by the poolside table — squeezable balloons that cushion pressure = the chemical BUFFERS that cushion sudden pH swings within seconds.

WHAT IT ENCODES

Buffering systems cushion pH changes within seconds — the body's first line of defense. Order/speed of defense: (1) chemical buffers (instant, bicarbonate + intracellular proteins/phosphate), (2) respiratory compensation (minutes to hours, alters PCO₂), (3) renal compensation (hours to 3-5 days, alters HCO₃ excretion/regeneration).

BOARD TRAP

WRONG answer is expecting full renal compensation within hours of an acute respiratory change. Renal HCO₃ adjustment takes 3-5 days to plateau. Discriminator: an acute respiratory acidosis shows minimal HCO₃ rise (~1 per 10 mmHg PCO₂) and a markedly low pH; a fully developed (chronic) one shows HCO₃ up ~3.5 per 10 mmHg with near-normal pH.

11. Bicarbonate soda box

WHAT YOU SEE

An orange 'BAKING SODA' box near the stand (it recurs around the scene) — baking soda = sodium bicarbonate = the BASE/alkali that raises pH.

WHAT IT ENCODES

Sodium bicarbonate = base (HCO₃), the metabolic alkali. Therapeutic role: IV NaHCO₃ is reserved for severe acidemia (generally pH <7.1), toxic-alcohol/salicylate poisoning (urinary alkalinization traps salicylate), hyperkalemia, and TCA overdose (sodium load narrows QRS). Each amp = 50 mEq HCO₃. Excess intake or volume contraction with maintained HCO₃ produces metabolic alkalosis.

BOARD TRAP

WRONG answer is bolusing bicarbonate for routine DKA or lactic acidosis with pH >7.1. In DKA bicarbonate is not recommended unless pH <6.9 — insulin + fluids clear ketones and regenerate HCO₃, and bicarbonate risks hypokalemia, paradoxical CNS acidosis, and cerebral edema in children. Discriminator: treat the underlying cause first; reserve HCO₃ for pH <6.9-7.1 or specific toxidromes.

12. Fan / blowing CO2 off

WHAT YOU SEE

Spinning fan blowing air

WHAT IT ENCODES

Ventilation moves CO₂ — the fast respiratory control of pH. Minute ventilation (rate × tidal volume) sets PCO₂: increased ventilation lowers PCO₂ and raises pH; hypoventilation does the reverse. This is the compensatory lever for primary metabolic disorders (Kussmaul respirations in DKA blow off CO₂ to compensate for metabolic acidosis).

BOARD TRAP

WRONG answer is interpreting Kussmaul (deep, rapid) breathing in a DKA patient as a primary respiratory alkalosis. It is appropriate respiratory COMPENSATION for the metabolic acidosis. Discriminator: apply Winter's formula — if measured PCO₂ matches 1.5×HCO₃ + 8 (±2), the low PCO₂ is expected compensation, not a separate respiratory disorder.